

ARTICLE 3

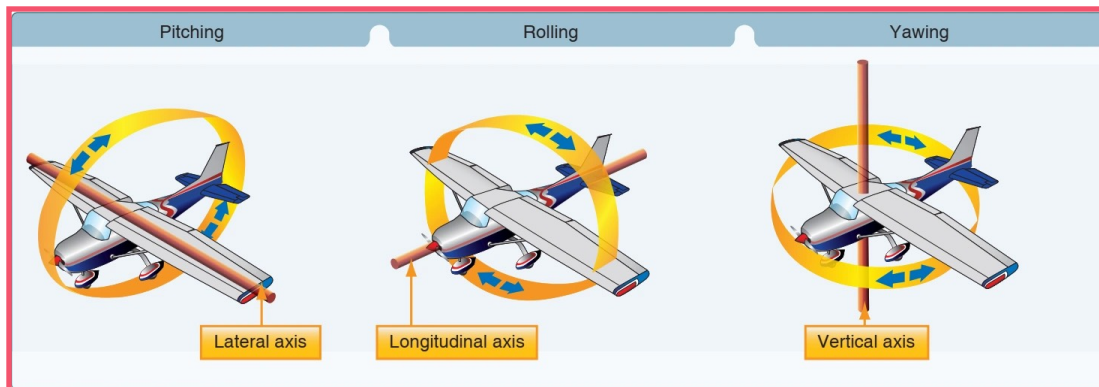
Aircraft stability & dutch roll phenomenon

What is aircraft stability ?

An airplane's stability is defined as its capacity to return to its original state following a disruption (pilot command or external effects like gusts).

When it comes to flying, there are three force components and three-moment components. The three forms of movement that an airplane may produce are roll, pitch, and yaw, as many of us already know.

The type of stability associated with the roll is lateral stability, whereas directional stability is associated with yaw. When a pilot wants to turn, she makes a roll movement; when she wants to change altitude, she makes a pitch movement; and when she wants to alter the direction of the aircraft's nose, she makes a yaw movement.



There are many forces acting upon an airplane . In stability state :

- The sum of all upward components of forces (not just lift) equals the sum of all downward components of forces (not just weight)
- The sum of all forward components of forces (not just thrust) equals the sum of all backward components of forces (not just drag)

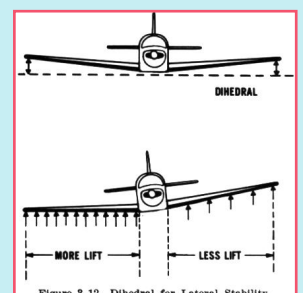
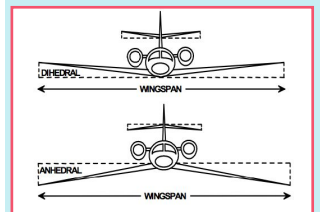
There are two general types of stability : **static and dynamic.**



Pape Mor NIANG

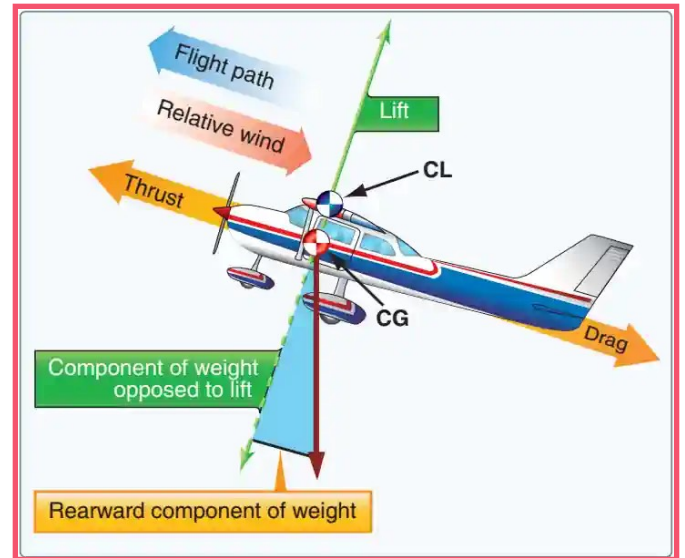
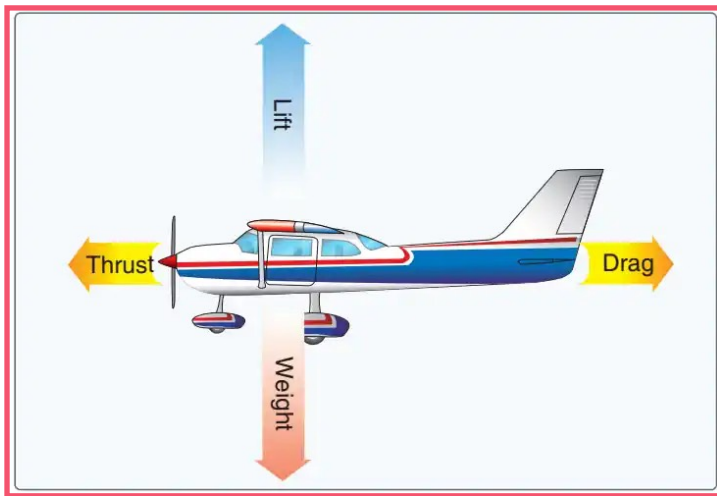
-Aerospace Engineer
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Wing configuration



Forward swept angle

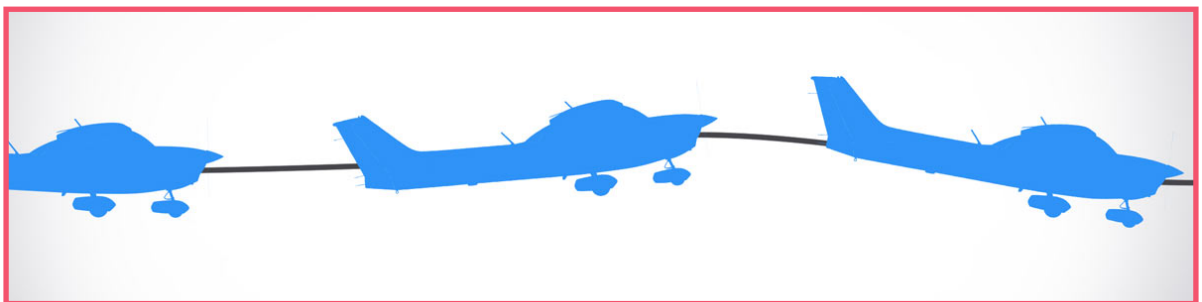




❖ **Static stability** : Static stability is the initial tendency of the aircraft once disturbed

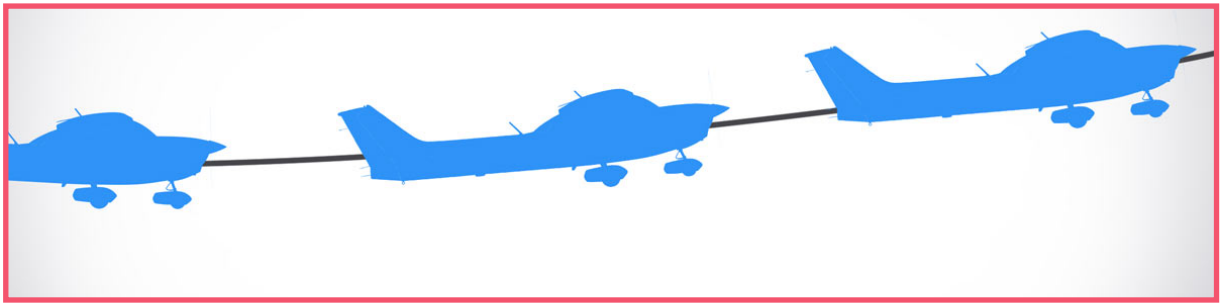
● **Positive Static Stability**

An aircraft that has positive static stability tends to return to its original attitude when it's disturbed. Let's say you're flying an aircraft, you hit some turbulence, and the nose pitches up. Immediately after that happens, the nose lowers and returns to its original attitude.



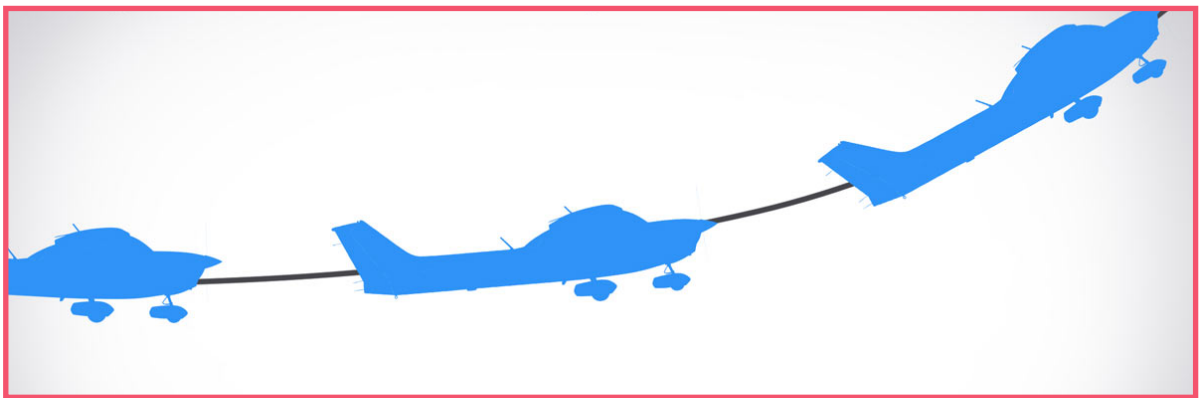
● **Neutral Static Stability**

An aircraft that has neutral static stability tends to stay in its new attitude when it's disturbed. For example, if you hit turbulence and your nose pitches up 5 degrees, and then immediately after that it stays at 5 degrees nose up, your airplane has neutral static stability.



● Positive Static Stability

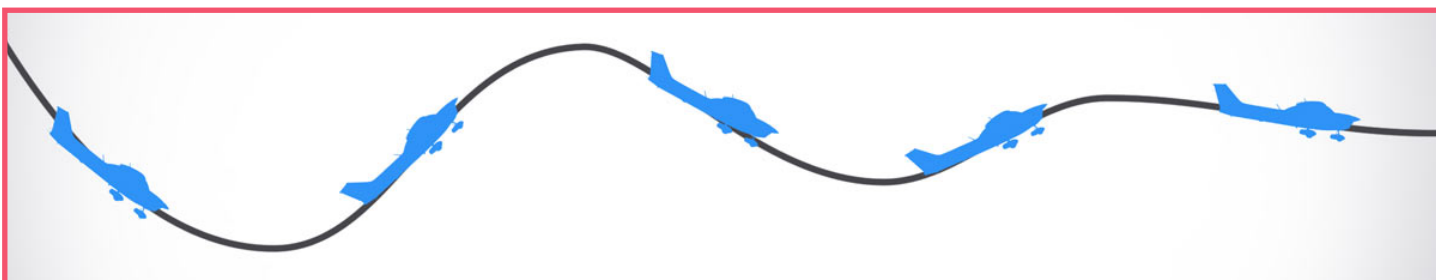
Finally, an aircraft that has negative static stability tends to continue moving away from its original attitude when it's disturbed. For example, if you hit turbulence and your nose pitches up, and then immediately continues pitching up, your airplane has negative static stability



❖ **Dynamic stability** : Dynamic stability is how an airplane responds over time to a disturbance. And it's probably no surprise that there are three kinds of dynamic stability as well :

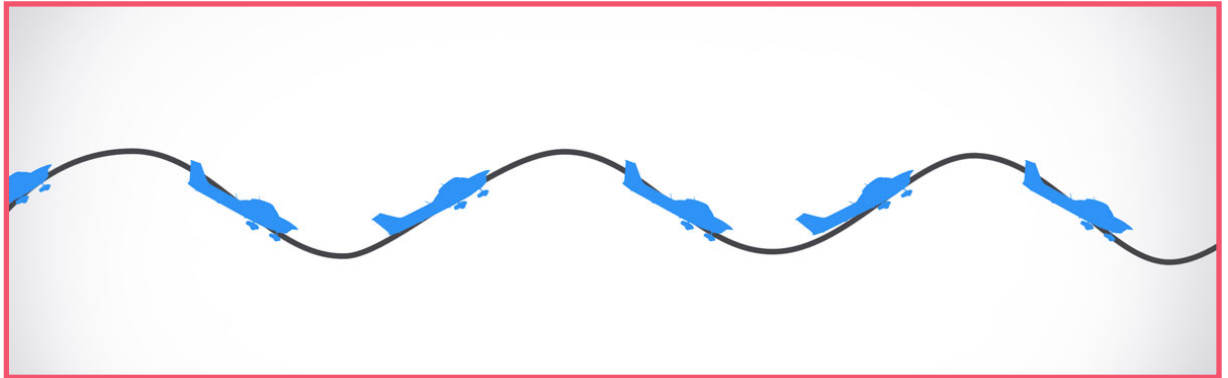
● Positive dynamic Stability

Aircraft with positive dynamic stability have oscillations that dampen out over time. The Cessna 172 is a great example. If your 172 is trimmed for level flight, and you pull back on the yoke and then let go, the nose will immediately start pitching down. Depending on how much you pitched up initially, the nose will pitch down slightly nose low, and then, over time, pitch nose up again, but less than your initial control input. Over time, the pitching will stop, and your 172 will be back to its original attitude.



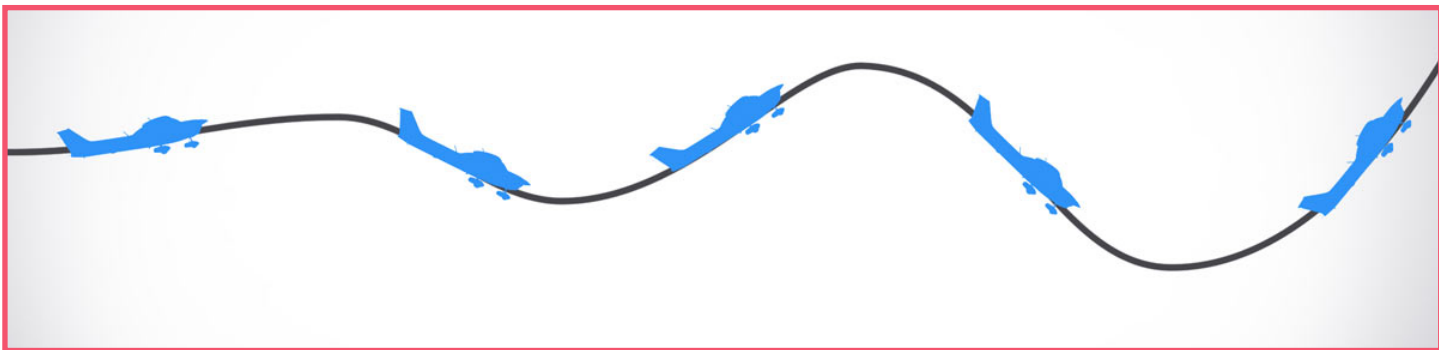
- **Neutral dynamic Stability**

Aircraft with neutral dynamic stability have oscillations that never dampen out. As you can see in the diagram below, if you pitch up a trimmed, neutrally dynamic stable aircraft, it will pitch nose low, then nose high again, and the oscillations will continue, in theory, forever.



- **Negative dynamic Stability**

Aircraft with negative dynamic stability have oscillations that get worse over time. The diagram below pretty much sums it up. Over time, the pitch oscillations get more and more amplified



- Longitudinal stability is the tendency of an aircraft to return to the trimmed angle of attack accomplished through elevators and rudders

Lateral and directional stability

❖ Lateral stability

Assume there is a disturbance in the plane, and it begins to sideslip, exactly as your pilot turned the plane right during your trip.

When an airplane sideslips to the right, its lateral stability creates an opposing moment that causes it to return to its original condition. *Wing dihedral, wing sweep, and the location of the wing* on the fuselage all contribute to this moment. High-wing airplanes, for example, will have superior stability but less mobility.

- ➔ **Wing Dihedral:** Dihedral refers to the angle formed by an aircraft's wings; positive dihedral indicates that the wings are pointed upwards. Positive dihedral allows the lower wing to produce more lift when the aircraft rolls, **resulting in a corrective movement**. The presence of positive dihedral improves lateral stability .
- ➔ **Wing Sweep:** Because of the swept, when an airplane is side sliding, the wing toward the sideslip will accept comparatively high-velocity airflow. Higher velocity will result in more lift, **resulting in a corrective movement** .
- ➔ **Wing Location:** Because of the position of the wings, high-wing airplanes **are more stable** .The center of gravity of high-wing planes lies under the wings, which improves static stability. General placement may be divided into three categories. Low-wing, mid-wing, and high-wing are the three types .

❖ Directional stability

On the other side, directional stability is linked to yaw. Although here, the stability is mostly provided by the vertical tail. As seen in the diagram, when an aircraft experiences a yaw disturbance caused by rudder commands or side winds, it rotates its nose but continues to fly in the same direction.

This indicates that the vertical tail has an angle of attack. This angle of attack will generate lift and cause the aircraft's nose to spin back with an opposite yaw moment in response to the disturbance.

Now that we've covered lateral and directional stability, we can move on to Dutch Roll. But before that let's talk about **slip** phenomenon .

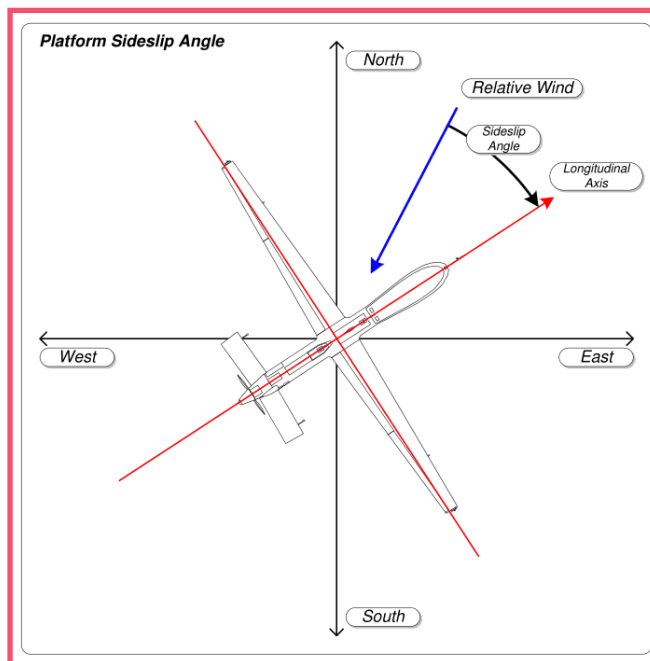
Slip motion ?

A **slip** is an aerodynamic state where an aircraft is moving *somewhat* sideways as well as forward relative to the oncoming airflow or relative wind. In other words, for a conventional aircraft, the nose will be pointing in the opposite direction to the bank of the wing(s). The aircraft is not in coordinated flight and therefore is flying inefficiently.

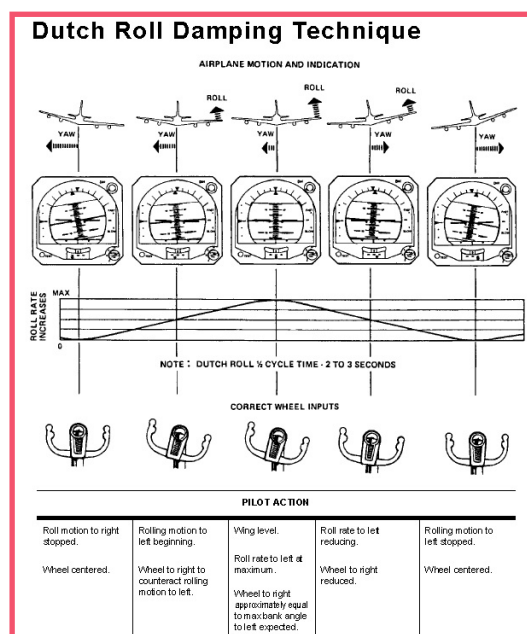
Forward slip ?

A flight maneuver in which the aircraft follows its intended flight path but its longitudinal axis is not aligned with this path.

The pilot applies the aileron in one direction and the rudder in the opposite. The aircraft then loses height rapidly. The greater the amount of bank and rudder application, the greater is the rate of descent. If the aircraft is displaced from the intended track and flown in such a way that it recovers its track, while maintaining its longitudinal axis parallel to the desired track, it is called a **sideslip**.



Finally dutch roll mode 🙄



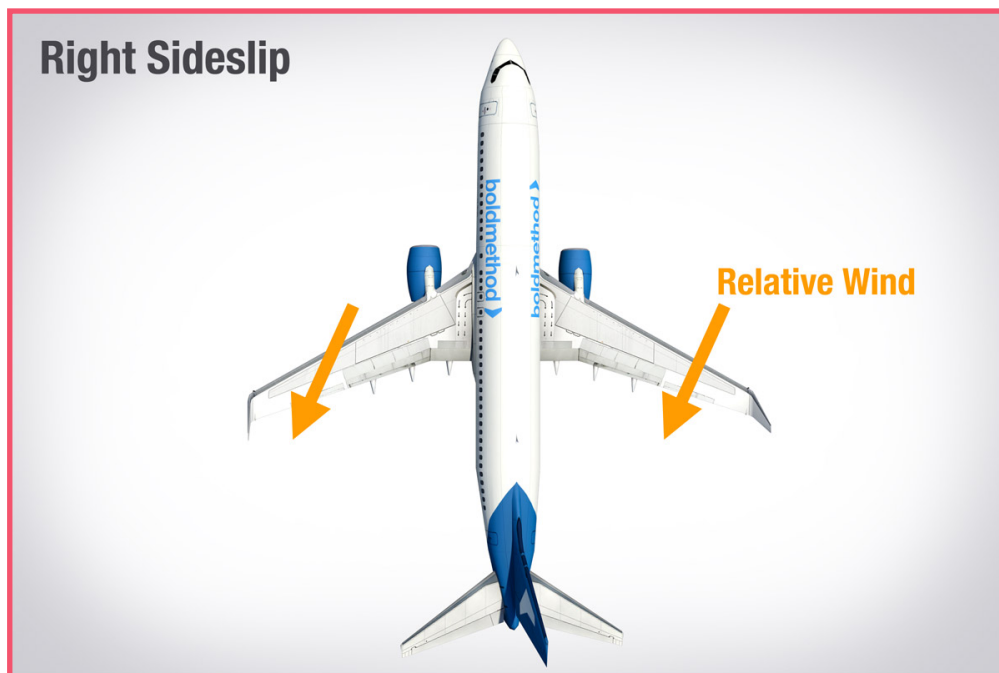
Dutch roll is a type of aircraft motion consisting of an out-of-phase combination of "tail-wagging" (yaw) and rocking from side to side (roll).

This yaw-roll coupling is one of the basic flight dynamic modes (others include phugoid, short period, and spiral divergence). This motion is normally well damped in most light aircraft, though some aircraft with well-damped Dutch roll modes can experience a degradation in damping as airspeed decreases and altitude increases. Dutch roll stability can be artificially increased by the installation of a yaw damper

Let's go back to the 737 example

Right slideslip ?

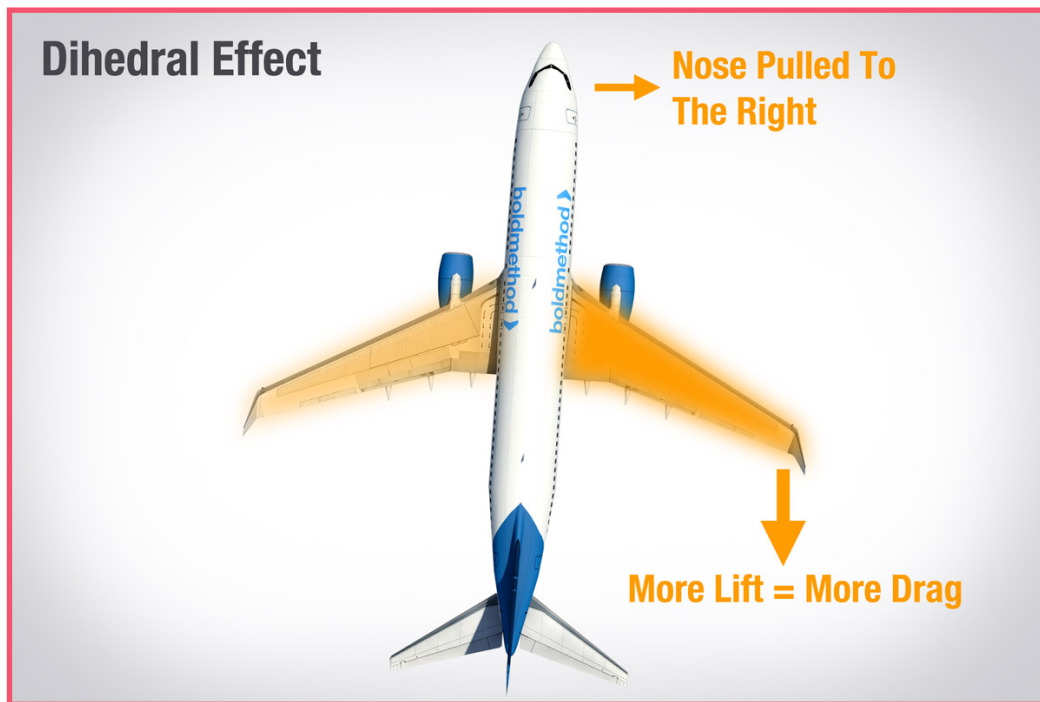
you've started an uncoordinated roll to the right. That tilts the lift vector to the right, causing a sideslip motion. Instead of the airflow blowing straight down your nose, it's now coming from the right.



Dihedral effect

you learned that only the air flowing parallel to the chord line creates lift. So, in the example above, the *right wing has more air flowing parallel to the chord line than the left wing, meaning the right wing generates more lift.*

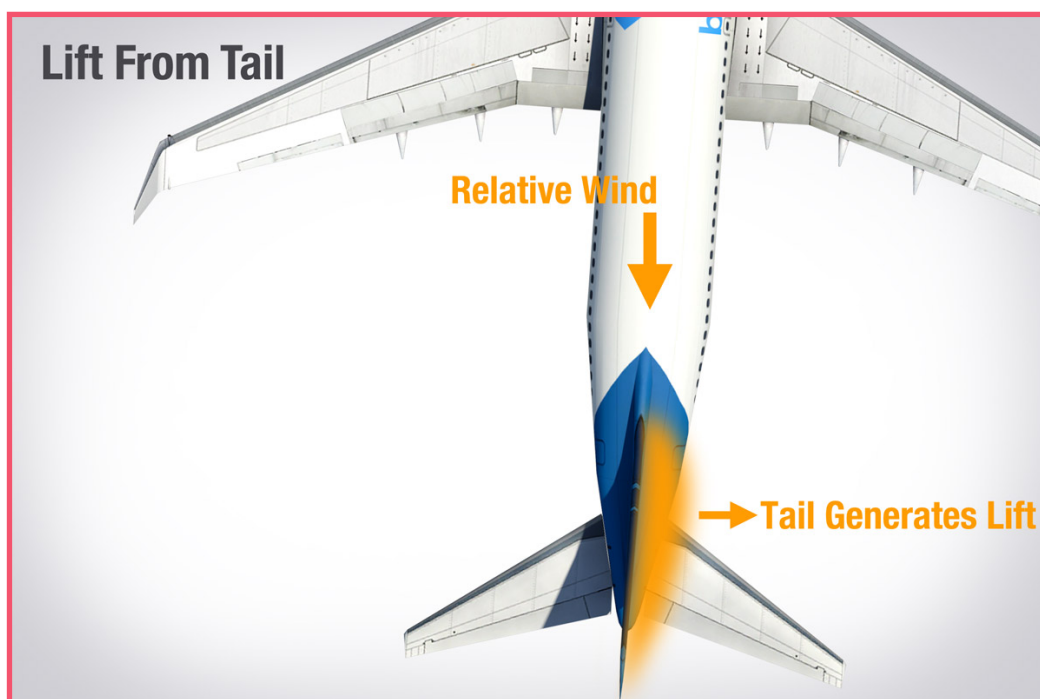
This effect is known as **dihedral effect** because it simulates the stabilizing effect of dihedral. It results in positive roll stability; the extra lift rolls the aircraft back towards level. However, this extra lift also generates drag that pulls your nose to the right.



Weaker Yaw Stability

Your vertical stabilizer helps stop the yaw motion caused by the right wing's drag. With your nose yawed to the right, the relative airflow approaches the vertical stabilizer from the left.

Your vertical stabilizer generates lift towards the right, yawing the nose back to the left.



Left sideslip

But, with a typical swept wing aircraft, **this yaw stability isn't as strong as the roll stability caused by the sweepback**. While the tail's still trying to line up the nose, the aircraft has over-banked to the left, causing a left sideslip. Now the sweepback starts to raise the left wing, rolling your 737 right. The drag from the left wing starts to pull the nose to the left.



MOST MODERN SWEEP WING AIRCRAFT HAVE YAW DAMPERS THAT AUTOMATICALLY CORRECT FOR DUTCH ROLL BY QUICKLY ADJUSTING THE RUDDER. IF YOUR YAW DAMPER'S INOPERATIVE, STOPPING THE ROLL CAN BE MORE TRICKY. MANY MODERN SWEEP-WING JETS WILL FLY THEMSELVES OUT OF DUTCH ROLL IF YOU STOP ADDING CONTROL INPUTS. HOWEVER, SOME OF THE OLDER JETS, LIKE THE 727, CAN BE DIFFICULT TO RECOVER. BUT WE'LL LEAVE THAT TECHNIQUE TO YOUR SIM INSTRUCTOR.